DS4420 Project - Tennis Serve Speed Predictor

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Introduction

Serve speed is a wildly variable metric in tennis, as it is influenced by a variety of other variables such as a player's physical attributes and form. We developed a bayesian and user-user CF model to attempt to predict serve speed. By understanding what metrics influence serve speed the most, it could help players understand more about their playstyle and increase their serve speed.

Motivation and Data

A good serve is one of the most impactful weapons in tennis, but players rarely know what serve speed is possible for their physical profile. We wanted to build models that take a player's height, weight, and age and return a predicted serve speed. This would give developing players a meaningful benchmark to train towards, based on the stats of real pros.

Our dataset consists of professional ATP players, with their average serve speed, height, weight, and age. After dropping players with missing physical measurements, we were left with approximately 100 players.

Methodology

User-User Collaborative Filtering

By finding players with similar physical metrics to the player we are trying to predict for, the CF will make predictions based on the similar players.

Bayesian Linear Regression

We fit a Bayesian linear regression model on player age, weight, and height, producing a distribution of predicted serve speeds.

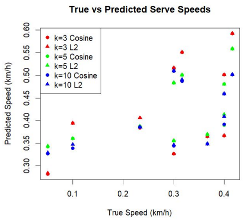
Results

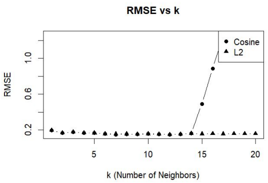
User-User Collaborative Filtering

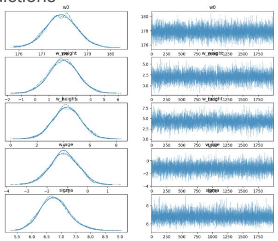
- Randomly masked 10 player's serve speeds to calculate RMSE, used k = 3, 5, and 10 (0.175/0.177, 0.165/0.165, 0.155/0.157) for cos/L2 respectively)
- MAEs were 0.146/0.148, 0.136/0.136, 0.126/0.127 for cos/L2 respectivly
- k=12 appears to be most optimal, after that cosine RMSE increases rapidly

Bayesian Linear Regression

- The model converged across all parameters (R-hat = 1.0)
- The intercept shows a baseline serve speed of 177.9 km/h
- Height was the strongest predictor (+4.3 km/h per SD), followed by weight (+2.1 km/h per SD), and age had a negative effect (-1.2 km/h per SD).
- Residual noise was 6.8 km/h, indicating meaningful uncertainty in predictions







Conclusion

- Not a lot of serve speed data easily accessible, so manual data collection may help with accuracy
- Humans are not very similar to each other, so the larger the sample size the better
- Serve speed is influenced by factors far beyond just height, weight and age. Things like mechanics and muscle composition play a role that physical attributes alone can't capture

Future Work

- Acquire more data from both genders, more metrics such as strength, arm length, etc.
- Use match-level data to explore how serve speed varies across different surfaces
- Model age non-linearly, since serve speed likely peaks around a players mid 20s
- Perhaps use a different model such as CNN to observe body mechanics
- Incorporate playing style or technique metrics, as different playing styles can play a role in serve speed

References

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Wong, F. K. H., J. H. Keung, N. M. Lau, D. K. Ng, J. W. Chung, and D. H. Chow. "Effects of Body Mass Index and Full Body Kinematics on Tennis Serve Speed." Journal of Human Kinetics 40 (2014): 21–28. doi:10.2478/hukin-2014-0003